Price Suppression and Reliability Shortfall in PJM's Capacity Market

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Abstract

The January 2025 Shapiro settlement modified PJM's capacity-auction demand curve to impose a price cap of approximately \$325 per megawatt-day of unforced capacity (UCAP) and a price floor of approximately \$175/MW-day on the 2026/27 and 2027/28 Base Residual Auctions, with a projected consumer benefit of over \$21 billion. Standard price-control theory predicts that a binding ceiling on a tight market truncates both the price signal and the quantity procured at the cap. I evaluate the connection by combining a calibrated symmetric supply function equilibrium of PJM's seven completed auctions with the Independent Market Monitor's unrestricted-VRR counterfactual. The SFE confirms that under IMM-reported concentration ($\mathrm{RSI}_3 < 1$) the unconstrained equilibrium clears at the design-level Point (a) cap and the settlement cap binds. The cap-incidence panel shows that a quantity shortfall relative to the reliability requirement emerged in both capped years (0.23% in 2026/27 and 4.7% in 2027/28), with no comparable shortfall in any prior auction in the sample. The IMM's mechanical counterfactual implies a system-level two-year revenue difference of \$13.08 billion and would have procured approximately 9,500 MW more capacity in 2027/28, against a Pennsylvania-stated figure of approximately \$21 billion that mixes the price and quantity channels. Under any reading consistent with real underlying scarcity, the cap induces a quantity shortfall and truncates the investment signal precisely in the years it would have been most informative.

1 Introduction

On January 28, 2025, Pennsylvania Governor Josh Shapiro announced an agreement with PJM Interconnection that capped capacity prices in the 2026/27 and 2027/28 Base Residual Auctions at approximately $325/MW-day of unforced capacity (UCAP) and floored them at approximately $175/MW-day (Office of Governor Josh Shapiro, 2025; Bresler and Keech, 2025). The settlement followed a Section 206 complaint filed against PJM's tariff one month earlier (Bresler and Keech, 2025) and was projected by the Commonwealth to save consumers over $21 billion across the two delivery years (Office of Governor Josh Shapiro, 2025).

Standard price-control theory delivers a sharp prediction. A binding ceiling on a market that would otherwise clear above the cap truncates supply at the cap; if the supply willing to clear at the cap is short of the planning target, the auction clears short. Weitzman (1974) formalizes the logic in the prices-vs-quantities framework, and Glaeser and Luttmer (2003) document a distinct allocative consequence: misallocation across heterogeneous demanders, qualitatively different from the standard deadweight triangle. The Shapiro cap, layered on top of a publicly known administrative demand curve and a concentrated supply, is a direct candidate for these predictions.

Both capped auctions cleared short of the reliability requirement. The 2026/27 BRA cleared 314 MW below the requirement (a 0.23% shortfall; 492 MW below the IRM-implied reserve target); the 2027/28 BRA cleared 6,623 MW below the requirement (a 4.7% shortfall; 6,517 MW below the IRM target). The PJM Independent Market Monitor (IMM) declared both auctions "not competitive" (Monitoring Analytics, LLC, 2026), and an IMM mechanical scenario that re-clears the same offers against the unrestricted (pre-settlement) VRR curve implies that approximately 9,500 MW more capacity would have cleared in 2027/28—enough to restore the reliability requirement plus a positive reserve margin. The shortage emerged when the cap began to bind, and not before.

This paper traces the connection. Section 4 states the textbook ceiling logic and uses the symmetric supply function equilibrium of Klemperer and Meyer (1989), Green and Newbery

(1992), and Holmberg (2008) to characterize the unconstrained equilibrium price; under the IMM-reported concentration ($RSI_3 < 1$), the equilibrium is at the cap and the settlement cap binds. Section 6 documents which administrative constraint was binding in each of the seven completed BRAs and shows that the shortage emerged in the same years the cap began to bind. Section 7 develops the dollar and physical consequences of the cap in each capped auction: in 2026/27 the cap suppressed price more than quantity; in 2027/28 it suppressed both. Section 8 reads the announced $21 billion savings claim in light of the shortfall and discusses the welfare and investment-signal implications of a binding cap during a tight-supply episode.

2 Literature Review

The supply function equilibrium concept, formalized by Klemperer and Meyer (1989) for demand-uncertainty oligopolies in which each firm commits to a price-quantity schedule before demand realizes, characterizes symmetric Nash equilibria by a first-order differential equation whose solutions form a continuum bounded above by the Cournot schedule and below by the competitive schedule $p = c$. Under demand uncertainty with bounded support, every trajectory between the two stationary solutions is itself a Nash equilibrium, so prediction and comparative statics require an equilibrium-selection criterion. Green and Newbery (1992) apply this framework to the England and Wales electricity market and establish the methodological template now standard in the calibrated SFE literature: derive the symmetric ODE, integrate backward from the capacity-constrained boundary, and calibrate to observed market structure. Holmberg (2008) shows that in capacity-constrained electricity markets the symmetric SFE is unique when rivals' capacity constraints simultaneously bind with positive probability, providing the boundary condition I use as an equilibrium-selection device in the deterministic BRA. Anderson and Hu (2008) and Holmberg (2007) extend the existence and characterization results to asymmetric firms; Rudkevich et al. (1998) bridges the canonical

Klemperer-Meyer theory and applied generation-market modeling.

Empirical and applied work has refined the SFE for electricity markets. Wolfram (1999) provides the canonical empirical test in the England and Wales duopoly and finds prices consistent with a strategic SFE rather than the competitive limit; Sweeting (2007) extends the analysis through restructuring. Sioshansi and Oren (2007) fits the SFE to ERCOT balancing-market data and assesses its predictive performance. Baldick et al. (2004) develop a tractable linear-affine variant useful for transmission-constrained networks, and Vasin et al. (2016) characterize the SFE specifically for uniform-price auction mechanisms of the kind PJM operates. Anderson and Xu (2005) is the most direct theoretical antecedent of the present paper: they derive supply function equilibria in spot markets with contracts and price caps and show that a binding cap pushes the equilibrium toward the capacity-cap boundary, with quantity rationed at the cap when supply is insufficient. The settlement studied here is one practical realization of that mechanism.

The capacity-market design literature originates from the "missing-money" problem: regulatory price caps in the wholesale energy market suppress the scarcity rents that would otherwise finance investment in peaking generation, leaving energy revenues insufficient to recover fixed costs of new entry (Joskow and Tirole, 2007; Joskow, 2008). Cramton and Stoft (2005) and Cramton and Ockenfels (2012) develop the design principles for a forward capacity instrument with a sloped administrative demand curve, and Cramton and Stoft (2007) defends uniform-price auctions over discriminatory alternatives in this setting. PJM's Reliability Pricing Model is analyzed dynamically by Hobbs et al. (2007), whose proposal informed the design of the VRR curve, and Bowring (2013) provides the institutional account from the Independent Market Monitor's perspective; Pfeifenberger et al. (2011) offers an early performance assessment. Hogan (2005) articulates the energy-only alternative, in which scarcity prices in the wholesale energy market are allowed to clear at administrative ceilings calibrated to the value of lost load and so deliver the investment signal that capacity-market advocates argue is otherwise missing.

The economics of binding price ceilings is foundational. Weitzman (1974) shows that the choice between price and quantity instruments depends on the relative slopes of marginal benefit and marginal cost; when supply is inelastic in the short run, price instruments dominate. Glaeser and Luttmer (2003) demonstrate, in the rent-control context, that binding ceilings generate a distinctive consumption misallocation across heterogeneous demanders, qualitatively different from the standard deadweight triangle. Experimental evidence in Vossler et al. (2009) confirms the qualitative shortage prediction in laboratory uniform-price wholesale-electricity markets. The empirical market-power literature on restructured electricity markets—Wolak (2003), Borenstein et al. (2002), and Bushnell et al. (2008)—supplies the methodological backdrop for distinguishing competitive from strategic clearing and motivates the cap-incidence diagnostic of Section 6. The applied question of this paper is the realization of these classical predictions in PJM's capacity market under the Shapiro settlement, with attention to the investment-signal cost of a binding cap during a tight-supply episode (Cramton and Stoft, 2005).

3 Institutional Background

PJM Interconnection operates the largest competitive wholesale electricity market in North America, coordinating about 201,000 MW of installed generating capacity across thirteen states and the District of Columbia. Its capacity market, the Reliability Pricing Model (RPM), was introduced in 2007 to procure forward commitments to be available during a future delivery year (Bowring, 2013; PJM Interconnection, L.L.C., 2025; Pfeifenberger et al., 2011). The market exists to address the "missing-money" problem: regulatory price caps in the wholesale energy market suppress the scarcity rents that would otherwise finance investment in peaking generation, leaving energy revenues insufficient to recover fixed costs of new entry (Joskow and Tirole, 2007; Joskow, 2008; Cramton and Stoft, 2005). RPM fills the gap with a forward instrument that pays for availability and so completes the revenue

stream needed for resource adequacy.

The primary procurement mechanism is the Base Residual Auction (BRA), a one-sided sealed-bid uniform-price auction held approximately three years before the start of the delivery year it procures. PJM acts as the administrative buyer through a publicly announced demand schedule (the Variable Resource Requirement, or VRR, curve) and aggregates resource-level offers into an upward-stepping supply schedule. Capacity is denominated in unforced capacity (UCAP) terms rather than nameplate installed capacity (ICAP). Each resource's accreditation ratio reflects the fraction of nameplate capacity expected to deliver during system stress, with forced-outage adjustments for thermal resources and Effective Load Carrying Capability (ELCC) adjustments for intermittent and energy-limited resources (PJM Interconnection, L.L.C., 2025, 2024). The reference dual-fuel combustion turbine used in PJM's planning parameters has an accreditation of approximately 79%, so a 100 MW nameplate unit clears as 79 MW of UCAP and is paid the clearing price on those 79 MW. Clearing is solved jointly across the RTO and constrained sub-LDAs subject to Capacity Emergency Transfer Limit (CETL) constraints, which bound the imports each LDA can receive from the rest of the system; when a CETL constraint binds, the LDA's clearing price separates from the RTO price and the constrained LDA pays a locational premium (PJM Interconnection, L.L.C., 2024).

A cleared resource receives the LDA's uniform clearing price, in dollars per MW-day of UCAP, for every day of the twelve-month delivery year, settled monthly. A 200 MW UCAP cleared resource at a $300/MW-day clearing price thus receives roughly $1.8 million per month in capacity revenue. The cleared resource also takes on a Capacity Performance (CP) obligation: it must be available to produce energy when PJM declares Performance Assessment Hours during the delivery year, typically tight intervals when reserves are low (PJM Interconnection, L.L.C., 2025). Failure to deliver during a Performance Assessment Hour triggers a non-performance charge calibrated to a multiple of Net CONE, and over-performance during the same hours earns a bonus payment funded from the non-performance

charges. The CP framework is intended to align capacity payments with delivery-year availability rather than mere ownership of nameplate capacity. The penalty mechanics themselves are not modeled in this paper; the SFE setup of Section 4 treats committed capacity as delivered.

Resources whose offers do not clear in the BRA have several alternatives. PJM conducts three Incremental Auctions (1st, 2nd, and 3rd IA) approximately twenty, ten, and four months before the delivery year, in which capacity needs that change after the BRA— driven by load-forecast revisions, planned-resource cancellations, or replacement-capacity transactions—are procured or released; a resource that did not clear the BRA can re-offer in any IA. Outside RPM, a load-serving entity may opt out of PJM's centralized procurement under the Fixed Resource Requirement (FRR) Alternative, securing capacity bilaterally from generators with which it contracts directly; FRR resources and load are removed from the BRA and bilateral capacity arrangements substitute for centralized clearing. A resource may also forgo capacity commitment for a delivery year and seek revenue solely from the energy and ancillary-service markets, accepting the lost capacity payment in exchange for operational flexibility. Finally, an existing resource contemplating retirement may file a Notice of Deactivation; if PJM determines that retirement would create a reliability concern that cannot be remedied through transmission upgrades or substitute resources, the unit may be designated Reliability Must-Run (RMR) and continue operating under a cost-of-service agreement at a regulated rate distinct from the BRA clearing price.

The VRR curve is the administrative demand schedule presented to sellers in each BRA (PJM Interconnection, L.L.C., 2025). Under the old three-point design (delivery years 2021/22 through 2025/26) it is parameterized by Point (a), the price cap at $1.5 \times$ Net CONE, Point (b) at $0.75 \times$ Net CONE at the reliability requirement, and Point (c) at zero price above the reliability requirement (Bowring, 2013; Pfeifenberger et al., 2011). The Net Cost of New Entry (Net CONE) is the annualized cost of a reference generating unit, net of projected energy and ancillary-service revenues, computed independently for each LDA (PJM Interconnection,

L.L.C., 2025, 2024). FERC approved a restructured four-point design taking effect with the 2026/27 delivery year (Federal Energy Regulatory Commission, 2025): an explicit price floor at approximately \$177–\$179/MW-day eliminates the collapse to zero price for surplus capacity. Both 2026/27 and 2027/28 cleared at exactly the new-design Point (a) price—which under the Shapiro/PJM settlement embodies the agreed maximum price (the settlement cap $\bar{p}_S$ described below)—motivating the analysis that follows. Recent BRAs have run on compressed lead times relative to the canonical three-year forward design (Table 1); the implications are discussed in Section 8.

Table 1: PJM Base Residual Auction Timeline and RTO Clearing Outcomes, 2021/22 to 2027/28

Delivery Year	BRA Date	Lead (mo.)	Clearing (\$/MW-day)	Net CONE (\$/MW-day)	VRR Design	At Cap
2021/22	May 2018	37.0	\$140.00	\$321.57	Old	No
2022/23	May 2021	12.5	\$50.00	\$260.50	Old	No
2023/24	July 2022	11.0	\$34.13	\$274.96	Old	No
2024/25	December 2022	17.8	\$28.92	\$293.19	Old	No
2025/26	July 2024	10.1	\$269.92	\$228.81	Old	No
2026/27	July 2025	10.2	\$329.17	\$212.14	New	Yes
2027/28	December 2025	17.7	\$333.44	\$242.52	New	Yes

Note: Prices in nominal dollars per MW-day. "At Cap" indicates the RTO clearing price equaled the Point (a) cap. Source: PJM BRA results and planning parameters.

PJM's primary market-power mitigation mechanism is the Three Pivotal Supplier (TPS) test. Supplier i is pivotal in LDA ℓ if removing i's capacity would cause aggregate supply to fall short of the reliability requirement. The Residual Supply Index for k suppliers is

$$\mathrm{RSI}_k^\ell \;=\; \frac{\text{Total capacity in LDA } \ell \;-\; \text{Capacity of top-}k \text{ sellers}}{\text{Reliability requirement for LDA } \ell}. \tag{1}$$

Mitigation triggers when $\mathrm{RSI}_1 \leq 1$ or $\mathrm{RSI}_3 \leq 1$, capping affected suppliers' offers at their

unit-specific Net Avoidable Cost Rate (Net ACR) (Bowring, 2013; Monitoring Analytics, LLC, 2026). Table 2 summarizes RTO-level outcomes: $RSI_3 < 1$ in every year of the sample (0.62–0.73), so the threshold is met throughout.

Table 2: TPS Test Results at the RTO Level: Residual Supply Indices by Delivery Year

Delivery Year	RSI_1	RSI_3	Participants	All Pivotal
2022/23	0.81	0.73	130	Yes
2023/24	0.78	0.68	134	Yes
2024/25	0.77	0.64	133	Yes
2025/26	0.82	0.62	128	Yes
2026/27	0.82	0.64	153	Yes
2027/28	0.82	0.63	155	Yes

Note: "All Pivotal" indicates $RSI_1 < 1$ for each of the three largest sellers individually. Source: PJM IMM State of the Market reports.

The high BRA clearing prices of 2025/26 and the anticipation of similarly elevated prices in subsequent auctions triggered a political intervention that directly modified the demand curve for two delivery years. On December 30, 2024, the Office of Governor Josh Shapiro filed a complaint against PJM's tariff with the Federal Energy Regulatory Commission under Section 206 of the Federal Power Act. The complaint argued that the VRR Point (a) price cap was set too high given lower expectations for new entry in the near term, and that compressed auction lead times limited the ability of new generation to respond to high prices in upcoming delivery periods, so high prices would not produce efficient entry and instead represented an unjust and unreasonable rate (Bresler and Keech, 2025).

Following confidential settlement discussions, PJM and the Governor's office announced an agreement in principle on January 28, 2025 (Office of Governor Josh Shapiro, 2025; Bresler and Keech, 2025). The agreement imposes a price band on the demand curves for both 2026/27 and 2027/28 delivery years: a price cap at approximately $325/MW-day UCAP

(denoted $\bar{p}_S$) and a price floor at approximately \$175/MW-day UCAP (denoted $\underline{p}_S$). The settlement does not impose an external price constraint on an otherwise unchanged auction; instead, it modifies the shape of the VRR demand curve itself. At prices above $\bar{p}_S$ the demand curve is flat (no capacity with offers above the cap can clear); between $\bar{p}_S$ and $\underline{p}_S$ the existing VRR sloped segment applies; at or below $\underline{p}_S$ the demand curve extends horizontally to clear all available capacity at the floor. This mechanism applies identically to the RTO and to every LDA demand curve, subject to CETL transmission limits (Bresler and Keech, 2025; Federal Energy Regulatory Commission, 2025). Throughout the paper, $\bar{p}$ denotes the pre-existing design-level Point (a) cap (1.5 × Net CONE, calibrated to investment incentives) and $\bar{p}_S$ denotes the lower settlement cap motivated by near-term consumer-cost concerns. Table 1 shows $\bar{p}$ = \$329.17/MW-day in 2026/27, only \$4.17/MW-day above $\bar{p}_S$ = \$325/MW-day; constrained-LDA Point (a) caps were larger in prior years, so the binding gap in those LDAs was correspondingly greater.

4 A Price-Cap Lens on the Shapiro Settlement

Standard price-control theory delivers a sharp prediction for a binding ceiling on a competitively cleared market. When the regulator imposes a cap $\bar{p}_S$ below the unconstrained clearing price p^*, demand at $\bar{p}_S$ exceeds the quantity sellers will provide; the gap between $D(\bar{p}_S)$ and $S(\bar{p}_S)$ is the shortage. Weitzman (1974) formalizes the choice between price and quantity instruments under uncertainty, and Glaeser and Luttmer (2003) document a distinct allocative consequence of binding ceilings: misallocation across heterogeneous demanders, qualitatively different from the standard deadweight triangle.

The Shapiro settlement applies this framework directly to PJM's Reliability Pricing Model. The settlement modifies the Variable Resource Requirement curve to be flat for all $p \geq \bar{p}_S$ = \$325/MW-day UCAP and adds a flat segment at $\underline{p}_S \approx$ \$175/MW-day for low prices (Section 3). The capped auction therefore clears whatever capacity sellers offer at or below

$\bar{p}_S$. If aggregate offers at the cap fall short of the reliability requirement q_b (the Point (b) quantity in the VRR curve), the auction clears at $(\bar{p}_S, Q_{\text{offered}})$ with $Q_{\text{offered}} < q_b$, and two consequences are visible separately: a truncated price signal on the upper segment, and—if supply is tight—a quantity shortfall against the planning target. Which of these channels was operative in 2026/27 and 2027/28 is the empirical question of Sections 6 and 7.

To distinguish a binding cap from a redundant one, the paper uses the symmetric supply function equilibrium (SFE) of Klemperer and Meyer (1989) as a characterization of the unconstrained equilibrium price. The setup follows the calibrated-electricity-market template of Green and Newbery (1992): K symmetric pivotal sellers with constant marginal cost c submit nondecreasing supply schedules $S_i(\cdot)$ against the publicly known VRR demand curve $D(\cdot)$, alongside a competitive fringe of size $\bar{Q}_f$ pricing at c. Equilibrium is selected by the Holmberg (2008) boundary condition, which requires each seller's supply function to attain its capacity at the design-level Point (a) cap, denoted $\bar{p}$ to distinguish it from the lower settlement cap $\bar{p}_S$. Under symmetry the equilibrium reduces to a scalar first-order ODE in the per-firm supply function $s(p)$, integrated backward from $s(\bar{p}) = \bar{q}$; the derivation is standard (Klemperer and Meyer, 1989; Green and Newbery, 1992; Rudkevich et al., 1998) and is not reproduced here.

The model's content for this paper is binary. The *at-cap regime* ($p^* = \bar{p}$) obtains whenever rivals' capacity constraints simultaneously bind, that is, when K pivotal sellers cannot meet residual demand at any interior price. The *interior regime* ($p^* < \bar{p}$) holds otherwise. The Shapiro cap $\bar{p}_S < \bar{p}$ binds in the at-cap regime by construction; in the interior regime it binds only if $\bar{p}_S < p^*$. Numerical solution of the symmetric ODE for the calibrations of Section 5 yields the at-cap regime for $K \leq 3$ and the interior regime for $K \geq 4$; the price-cost margin falls by roughly 20–25 percentage points across this regime change.[1]

PJM's auction concentration is consistent with the $K \leq 3$ regime. The IMM reports

[1]For 2026/27 with $c = \$150$/MW-day, the symmetric SFE clears at the cap ($p^* = \bar{p} = \$329$, $L = 0.54$) under $K = 3$ and at $p^* \approx \$228$ ($L = 0.34$) under $K = 4$. The qualitative pattern is the same for the other six calibration years.

$\text{RSI}_3 < 1$ in every sample year (Table 2), meaning the three largest sellers are collectively pivotal at the reliability requirement. A symmetric $K = 3$ calibration is therefore the natural choice and is maintained throughout. The SFE plays a focused role in what follows: it identifies which auctions feature a binding settlement cap, and it supplies an unconstrained-market counterfactual price against which the cap's effect on cleared MW can be measured in Section 7. In the at-cap regime, the SFE matches the observed clearing price by construction, not as an independent empirical test: its content for this paper is the regime classification itself, with the dollar value of $\bar{p}$ read directly from PJM's planning parameters.

5 Data and Calibration

The calibration draws on four publicly available datasets. PJM publishes VRR curve anchor points, Net CONE, CETL, and reliability requirements in advance of each BRA, and clearing prices and quantities by LDA after each auction. The Independent Market Monitor (IMM) reports HHI, Residual Supply Indices, and Capacity Performance offer-cap composition in its annual State of the Market reports. I join the four datasets at the LDA-delivery-year level into a master panel covering all seven completed BRAs, 2021/22 through 2027/28.

Table 3 reports the RTO VRR parameters. Under the old three-point design, Points (a) and (b) equal 1.5× and 0.75× Net CONE respectively, with a transitional administrative cap in 2025/26. Under the new four-point design (2026/27 onward), Points (b) and (c) collapse to the floor price of \$177–\$179/MW-day and Point (a) is set by FERC Order ER25-1357.

Table 3: RTO VRR Curve Parameters, 2022/23–2027/28

Delivery Year	Net CONE ($/MW-day)	Pt. (a) Price	Pt. (a) MW	Pt. (b) Price	Pt. (b) MW	VRR Design
2022/23	$260.50	$390.75	136,076	$195.38	139,656	Old
2023/24	$274.96	$412.44	135,914	$206.22	139,473	Old
2024/25	$293.19	$439.79	138,341	$219.89	141,910	Old
2025/26	$228.81	$451.61	133,554	$171.61	137,160	Old
2026/27	$212.14	$329.17	134,887	$177.24	136,361	New
2027/28	$242.52	$333.44	141,681	$179.55	143,218	New

Note: Prices in nominal dollars per MW-day; quantities in MW. Pt. (b) MW is the reliability requirement. Source: PJM Planning Parameters.

I set $K = 3$ symmetric strategic sellers based on the IMM's Residual Supply Index. Across the six sample years with published IMM RSI data (2022/23 onward; the 2021/22 BRA predates the available IMM panel), $\text{RSI}_3 < 1$ at the RTO level (Table 2), meaning the three largest sellers are collectively pivotal: no competitive outcome is supportable without their participation. RTO HHI exceeds 2,500 in every sample year, with most constrained LDAs above 4,000—highly concentrated by Department of Justice benchmarks. Sensitivity of the at-cap conclusion to K is bounded by Section 4: at $K = 4$ the at-cap regime breaks.

The constant marginal cost is set to $c = \$150$/MW-day, the combined-cycle sector average reported in the 2025 IMM State of the Market (Monitoring Analytics, LLC, 2026, Table 7-38, $149.32, rounded). The fringe is modeled as an inelastic block offered at c with capacity $\bar{Q}_f$ equal to total installed capacity less the capacity of the three largest sellers. The model is calibrated and solved for all seven completed BRA delivery years.

6 Cap-Incidence and the Emergence of Shortage

This section documents which administrative constraint was binding in each of PJM's seven completed Base Residual Auctions and shows that a quantity shortfall against the reliability requirement emerged precisely when the Shapiro settlement cap began to bind. The textbook prediction of Section 4—that a binding ceiling on a tight market produces a quantity shortfall against the planning target—is matched by the timing of the observed shortfall in PJM.

Table 4: Cap-Incidence Panel: Which Administrative Constraint Was Binding, PJM Base Residual Auctions 2021/22–2027/28 (RTO Level)

Year	VRR Design	Clearing price	Net ACR	VRR Pt (a)	Settle. cap	Reserve excess IRM (MW)	TPS mitig.	Binding cap
2021/22	Old	$140	$150	$482	—	—	—	None
2022/23	Old	$50	$150	$391	—	—	0.0%	None
2023/24	Old	$34	$150	$412	—	+5,687	7.3%	TPS active
2024/25	Old	$29	$150	$440	—	+3,278	2.2%	TPS active
2025/26	Old	$270	$150	$452	—	-205	5.5%	TPS active
2026/27	New	$329	$150	$329	$325	-492	6.3%	Settlement
2027/28	New	$333	$150	$333	$325	-6,516	1.0%	Settlement

Notes: Prices in nominal $/MW-day UCAP. Net ACR is the IMM-published sector-average Avoidable Cost Rate ($149.32, rounded to $150) used as the default mitigation offer cap for resources subject to the Three Pivotal Supplier (TPS) test. VRR Pt (a) is the design-level price cap on the unmodified Variable Resource Requirement curve. Settlement cap is the Shapiro/PJM $325/MW-day UCAP cap that took effect for the 2026/27 and 2027/28 delivery years. Reserve excess IRM (MW) is committed UCAP minus IRM-implied required reserves (negative = shortfall). TPS mitigation percentage is the share of generation resources that submitted Capacity Performance offers and had unit-specific Market Seller Offer Caps calculated by the IMM. Binding cap classification: "Settlement" = cleared within 2% of the new-design Point (a), which under the Shapiro/PJM agreement embodies the settlement cap; "TPS active" = TPS mitigation calculated for $\geq 1\%$ of resources with clearing more than 5% below Point (a), indicating mitigation operative for some marginal resources but not pinning headline clearing; "None" = clearing well below all administrative caps with TPS mitigation calculated for $< 1\%$ of resources. Sources: PJM BRA results (clearing prices, cleared MW, reliability requirement); IMM State of the Market Reports 2021–2025, Section 5 Capacity (TPS mitigation counts, reserve-margin panel). The 2021/22 BRA (held May 2018) predates locally-available IMM SoMs, so TPS mitigation and reserve-excess data are reported as "—".

Three regimes are visible in Table 4, distinguished by the joint pattern of clearing price, reserve excess relative to the Installed Reserve Margin (IRM), and IMM-applied mitigation activity.

The first regime, covering 2021/22 through 2024/25, is one of comfortable supply margins and cost-anchored clearing. Reserve excess IRM was strongly positive in 2023/24 and 2024/25 (+5,687 MW and +3,278 MW; data unavailable for 2021/22 and 2022/23). Clearing prices ranged from \$29 to \$140/MW-day, well below both the sector-average Net ACR and the design-level VRR Point (a) cap of \$391–\$482/MW-day. TPS mitigation was applied to 0–7% of resources, but with the marginal cleared offer well below any active cap, mitigation operative for a minority of supra-marginal resources did not pin the clearing outcome.

The second regime, in 2025/26, is a transition. Reserve excess IRM turned slightly negative for the first time (−205 MW), the clearing price jumped from \$29 to \$270/MW-day, and TPS mitigation rose to 5.5% of resources. Yet \$270 remains far below the design Point (a) of \$452, so no administrative cap was binding the clearing. 2025/26 is the auction in which the underlying supply–demand balance tightened materially without any cap yet binding the headline outcome, and—importantly—the auction still procured adequate capacity.

The third regime, comprising 2026/27 and 2027/28, is one of cap-binding clearing under negative reserve excess. Both auctions cleared at the new-design VRR Point (a) value, which under the Shapiro/PJM settlement embodies the agreed maximum price. Reserve excess IRM was −492 MW in 2026/27 and −6,517 MW in 2027/28; the IMM declared both auctions "not competitive" (Monitoring Analytics, LLC, 2026). The drop in TPS mitigation application from 6.3% in 2026/27 to 1.0% in 2027/28 is consistent with the settlement cap moving sufficiently close to the underlying offer distribution that fewer individual offers exceeded their unit-specific Net ACR caps. The shortfall relative to the reliability requirement *emerged in the same years that the cap became binding*, not before.

The IMM-published distribution of resources across offer-cap categories is consistent with the cap-incidence reading. Across the sample, Capacity Performance offers fall predominantly into two clusters: price-takers offering at \$0 (12–35% of resources) and resources at their cost-basis cap (57–81%), together accounting for 88–93% of resources in every year. This bimodality is a structural feature of the mitigation rules, not a behavioral choice: the Default

ACR offer cap is a hard ceiling for any resource subject to mitigation, and the price-taker option is the only alternative for resources unwilling to compute and defend a unit-specific cap. The settlement cap, when binding, truncates the upper cluster: a resource whose cost-basis cap exceeds $325 cannot clear above $325 even if its unit-specific calculated cap would otherwise allow it. Whether the settlement induced *additional* clustering near $325/MW-day cannot be determined from the aggregated IMM categories; unit-level offer histograms will not be public until 2029.

The cap-incidence panel offers a direct answer to a recurring question about the post-settlement auctions: did seller behavior change between the slack and tight regimes, or did the binding administrative constraint change while the underlying competitive process remained similar? The data favor the latter interpretation. In slack years, clearings of $29–$140/MW-day fell well below any plausible unconstrained strategic price (cost-anchored offers and TPS mitigation pinned the headline price). In the cap-binding years, clearing equaled the cap by construction in any model in which the at-cap regime obtains. The dramatic price changes between regimes therefore reflect, at least in part, a change in which administrative constraint binds, not a structural shift in conduct. The independent evidence that this is not a circular reading is the *timing* of the reserve-margin flip: reserve excess IRM turned negative in the 2025/26 transition auction—one year before the cap took effect—and stayed negative in the two cap-binding years that followed.

This reading does not contradict the IMM's substantive assessment. The IMM attributes the underlying scarcity in 2026/27 and 2027/28 to data center load growth rather than to any change in seller behavior or to the settlement itself (Monitoring Analytics, LLC, 2026). The contribution of this section is to make that interpretation visible at a glance and to frame the dollar consequences—developed for the two capped auctions in Section 7—as the result of overlaying a binding administrative cap on a market entering scarcity, rather than as the result of strategic conduct that the cap then disciplined.

7 Capped Auctions: Quantity Rationing in 2026/27 and 2027/28

The cap-incidence panel of Section 6 establishes that a quantity shortfall against the reliability requirement emerged in the two delivery years for which the Shapiro settlement cap was binding. This section turns to the dollar and physical consequences of the shortfall in each auction. Both cases use the IMM's mechanical unrestricted-VRR counterfactual—which re-clears the same offers against the pre-settlement demand curve—as the principal benchmark.

The 2026/27 BRA cleared at $329.17/MW-day, which under the new VRR design embodies the settlement cap. The market cleared 314 MW below the RTO reliability requirement of 134,519.5 MW (a 0.23% shortfall) and 492 MW below the IRM-implied reserve target; the IMM declared the auction "not competitive" (Monitoring Analytics, LLC, 2026). Capacity Performance offers had unit-specific Market Seller Offer Caps calculated for 6.3% of generation resources.

The IMM's mechanical counterfactual scenario re-clears the same offers against the unrestricted (pre-settlement) VRR demand curve and yields total revenues of $19.29 billion, a 19.7% increase over the $16.12 billion actual revenue (Monitoring Analytics, LLC, 2026). The system-level revenue increase substantially exceeds what the $4.17/MW-day RTO-level binding gap alone implies; most of the difference comes from constrained Locational Deliverability Areas in which local Point (a) caps lie further above the settlement cap than at the RTO level, so re-clearing those LDAs against the unrestricted VRR raises both their local prices and the imports they pay for. With the RTO reserve margin only marginally negative, the cap truncated the price signal more than it truncated the quantity procured at the system level.

The 2027/28 BRA cleared at $333.44/MW-day, again at the new-design Point (a). The market cleared 6,623 MW below the RTO reliability requirement of 141,101 MW—a 4.7% shortfall that the IMM characterized as material; the auction was declared "not competitive"

on the same basis as 2026/27. Unit-specific MSOCs were calculated for only 1.0% of resources, consistent with the settlement cap moving close enough to the underlying offer distribution that fewer individual offers exceeded their cost-basis caps.

The IMM unrestricted-VRR counterfactual produces total revenues of $26.32 billion, a 60.4% increase over the $16.41 billion actual revenue. The disproportionately large increase relative to 2026/27 reflects two compounding channels: a higher counterfactual price (the unrestricted VRR allows clearing further up the demand curve) and a higher counterfactual cleared quantity (offers above the settlement cap that were excluded under the actual auction would have cleared). A back-of-envelope decomposition: at a counterfactual price of $500/MW-day and the actual cleared quantity of 134,478 MW over 366 days, revenues would be approximately $24.6 billion; the IMM's $26.32 billion exceeds this by roughly $1.7 billion, implying a counterfactual cleared quantity of approximately 144,000 MW—about 9,500 MW above actual and roughly equal to the reliability requirement plus a positive reserve margin. Under the IMM scenario the unrestricted auction would have realized higher prices and procured adequate capacity; under the cap, the auction achieved neither.

Table 5: Capacity Market Revenue Under Counterfactual Cap Regimes (System Total)

Scenario	2026/27 revenue	2027/28 revenue	Two-year total	Implied savings
Actual (settlement-modified VRR)	\$16.12B	\$16.41B	\$32.53B	—
IMM unrestricted-VRR counterfactual	\$19.29B	\$26.32B	\$45.62B	\$13.08B
Percent increase vs. actual	+19.7%	+60.4%	+40.2%	—
\$500/MW-day at actual cleared MW	\$24.51B	\$24.60B	\$49.10B	\$16.57B
Stated Pennsylvania claim	—	—	∼\$21B	∼\$21B

Notes: Revenues in nominal dollars over the twelve-month delivery year (366 days for 2027/28). The IMM unrestricted-VRR counterfactual holds offers fixed and re-clears against the pre-settlement VRR demand curve; figures from the IMM 2025 State of the Market Report (Monitoring Analytics, LLC, 2026). The \$500/MW-day scenario multiplies the actual cleared UCAP (134,205 MW in 2026/27, 134,478 MW in 2027/28) by Pennsylvania's stated counterfactual price; this scenario corresponds to the Pennsylvania counterfactual but holds quantity at the capped level rather than re-clearing demand. The \$21 billion claim (Office of Governor Josh Shapiro, 2025) reflects additional revenue contributions from constrained Locational Deliverability Areas where local Point (a) prices exceeded \$325/MW-day in prior years. Implied savings are differences between the counterfactual scenario and the actual outcome.

The 2027/28 case is the cleanest evidence in the sample of the quantity-rationing channel of a binding administrative cap. The 2026/27 case isolates the price channel in a market only marginally short of adequacy. Together the two cases trace the empirical realization of the textbook prediction of Section 4: a binding ceiling truncates the price signal in any tight market, and where supply at the cap is insufficient to meet the planning target, it also truncates the quantity procured.

8 Conclusion

Standard price-control theory predicts that a binding ceiling on a tight market truncates the price signal and produces a quantity shortfall against the planning target. PJM's two capped delivery years, 2026/27 and 2027/28, show both effects: clearing prices pinned at the new-design Point (a) value and cleared capacity short of the reliability requirement by 314 MW and 6,623 MW, respectively (Section 7). The shortfall emerged in the same years the cap became binding (Section 6), with no comparable shortfall in any prior auction in the sample. The textbook prediction and the empirical realization line up.

The Pennsylvania-stated savings figure of approximately $21 billion compares actual revenue under the settlement-modified VRR to a counterfactual price "over $500/MW-day" (Office of Governor Josh Shapiro, 2025) applied to the capped quantity, and so mixes the price and quantity channels. The IMM's mechanical unrestricted-VRR counterfactual, which holds offers fixed and re-clears against the pre-settlement demand curve, implies two-year revenues of $45.62 billion against actual $32.53 billion—a system-level difference of $13.08 billion—and would have procured approximately 9,500 MW more capacity in 2027/28 (Table 5). Treating the difference as "savings" conflates a price reduction with a quantity-rationing shortfall: under the cap, consumers paid less, and—in 2027/28 particularly—the auction procured less than the reliability target.

Lead-time compression amplifies the investment-signal cost. The 2025/26, 2026/27, and 2027/28 BRAs were held with 10–18 months of lead time against a canonical three-year forward design (Table 1). A 10-month horizon is too short for greenfield gas but non-trivial for demand response, battery storage, and capacity upgrades to existing plants; auction prices inform retirement-deferral decisions on the same timescale (Cramton and Stoft, 2005). A two-year cap during a tight-supply episode therefore disrupts the retirement and short-deployment signal precisely in the years when it would have been most informative, with the new four-point VRR design potentially recovering some of that signal in subsequent auctions (Federal Energy Regulatory Commission, 2025).

The welfare reading depends on whether the underlying scarcity is real or strategic. If sellers were exercising market power and the unconstrained equilibrium reflected rents above efficient scarcity prices, the cap is unambiguously welfare-improving. If the underlying scarcity is real—driven by load growth and supply tightening, which the IMM attributes to data-center demand (Monitoring Analytics, LLC, 2026)—the cap induces a quantity shortage in years when capacity adequacy is most needed and truncates the very price signal the capacity market exists to send. The cap-incidence panel of Section 6, the declared "not competitive" status of both auctions, and the realization of the textbook ceiling-induced shortfall together favor the latter reading.

The paper is restricted to evaluating the connection between the standard price-control prediction and the PJM evidence. Two natural extensions for distinguishing scarcity from conduct quantitatively are an asymmetric-firm SFE that recovers seller-specific costs from offer data, and a dynamic capacity-adequacy model connecting markups to entry and retirement decisions.

References

Edward J. Anderson and Xinmin Hu. Finding supply function equilibria with asymmetric firms. *Operations Research*, 56(3):697–711, 2008.

Edward J. Anderson and Huifu Xu. Supply function equilibrium in electricity spot markets with contracts and price caps. *Journal of Optimization Theory and Applications*, 124(2): 257–283, 2005.

Ross Baldick, Ryan Grant, and Edward Kahn. Theory and application of linear supply function equilibrium in electricity markets. *Journal of Regulatory Economics*, 25(2):143–167, 2004.

Severin Borenstein, James B. Bushnell, and Frank A. Wolak. Measuring market inefficiencies in California's restructured wholesale electricity market. *American Economic Review*, 92 (5):1376–1405, 2002.

Joseph Bowring. Capacity markets in PJM. *Economics of Energy & Environmental Policy*, 2 (2):47–64, 2013.

Stu Bresler and Adam Keech. Consultation: Capacity market demand curve adjustments pursuant to proposed settlement. Special MC/TOA-AC Presentation, PJM Interconnection, February 2025. February 7, 2025.

James B. Bushnell, Erin T. Mansur, and Celeste Saravia. Vertical arrangements, market structure, and competition: An analysis of restructured US electricity markets. *American Economic Review*, 98(1):237–266, 2008.

Peter Cramton and Axel Ockenfels. Economics and design of capacity markets for the power sector. *Zeitschrift für Energiewirtschaft*, 36(2):113–134, 2012.

Peter Cramton and Steven Stoft. A capacity market that makes sense. *Electricity Journal*, 18(7):43–54, 2005.

Peter Cramton and Steven Stoft. Why we need to stick with uniform-price auctions in electricity markets. *Electricity Journal*, 20(1):26–37, 2007.

Federal Energy Regulatory Commission. PJM Interconnection, L.L.C. Docket No. ER25-1357, Order Accepting Tariff Revisions, 2025. Accepting PJM compliance filing for Capacity Market Demand Curve Adjustments Pursuant to Settlement.

Edward L. Glaeser and Erzo F. P. Luttmer. The misallocation of housing under rent control. *American Economic Review*, 93(4):1027–1046, 2003.

Richard J. Green and David M. Newbery. Competition in the British electricity spot market. *Journal of Political Economy*, 100(5):929–953, 1992.

Benjamin F. Hobbs, Ming-Che Hu, Javier G. Iñón, Steven E. Stoft, and Murty P. Bhavaraju. A dynamic analysis of a demand curve-based capacity market proposal: The PJM reliability pricing model. *IEEE Transactions on Power Systems*, 22(1):3–14, 2007.

William W. Hogan. On an "energy only" electricity market design for resource adequacy. Technical report, John F. Kennedy School of Government, Harvard University, September 2005.

Pär Holmberg. Supply function equilibrium with asymmetric capacities and constant marginal costs. *Energy Journal*, 28(2):55–82, 2007.

Pär Holmberg. Unique supply function equilibrium with capacity constraints. *Energy Economics*, 30(1):148–172, 2008.

Paul L. Joskow. Capacity payments in imperfect electricity markets: Need and design. *Utilities Policy*, 16(3):159–170, 2008.

Paul L. Joskow and Jean Tirole. Reliability and competitive electricity markets. *RAND Journal of Economics*, 38(1):60–84, 2007.

Paul D. Klemperer and Margaret A. Meyer. Supply function equilibria in oligopoly under uncertainty. *Econometrica*, 57(6):1243–1277, 1989.

Monitoring Analytics, LLC. 2025 state of the market report for PJM. Annual Report, Independent Market Monitor for PJM, 2026.

Office of Governor Josh Shapiro. Governor Josh Shapiro reaches agreement with PJM to prevent unnecessary price hikes and save consumers over $21 billion on utility bills. Commonwealth of Pennsylvania Press Release, January 2025. Accessed March 2026. Available at https://www.pa.gov/governor/newsroom/2025-press-releases/gov-shapiro-agreement-pjm-prevent-price-hikes-save-consumers-ove.

Johannes P. Pfeifenberger, Kathleen Spees, Kevin Carden, and Nicholas Wintermantel. Second performance assessment of PJM's reliability pricing model. Technical report, The Brattle Group, 2011. Prepared for PJM Interconnection.

PJM Interconnection, L.L.C. *PJM Manual 20A: PJM Resource Adequacy Analysis*. PJM Interconnection, 2024.

PJM Interconnection, L.L.C. *PJM Manual 18: PJM Capacity Market*. PJM Interconnection, 2025.

Aleksandr Rudkevich, Max Duckworth, and Richard Rosen. Modeling electricity pricing in a deregulated generation industry: The potential for oligopoly pricing in a poolco. *Energy Journal*, 19(3):19–48, 1998.

Ramteen Sioshansi and Shmuel Oren. How good are supply function equilibrium models: An empirical analysis of the ERCOT balancing market. *Journal of Regulatory Economics*, 31 (1):1–35, 2007.

Andrew Sweeting. Market power in the England and Wales wholesale electricity market 1995–2000. *Economic Journal*, 117(520):654–685, 2007.

Alexander Vasin, Marina Dolmatova, and Gerhard-Wilhelm Weber. Supply function equilibria for uniform price auction in oligopolistic markets. *Central European Journal of Operations Research*, 24(4):819–831, 2016.

Christian A. Vossler, Timothy D. Mount, Robert J. Thomas, and Ray D. Zimmerman. An experimental investigation of soft price caps in uniform price auction markets for wholesale electricity. *Journal of Regulatory Economics*, 36(1):44–59, 2009.

Martin L. Weitzman. Prices vs. quantities. *Review of Economic Studies*, 41(4):477–491, 1974.

Frank A. Wolak. Measuring unilateral market power in wholesale electricity markets: The California market, 1998–2000. *American Economic Review*, 93(2):425–430, 2003.

Catherine D. Wolfram. Measuring duopoly power in the British electricity spot market. *American Economic Review*, 89(4):805–826, 1999.